

Found Your World

A verified path from model limits to a World Bible v0.1

Computer Science II • Dr. Jeremy P. Evert

August 31, 2026

Today's destination

TODAY'S DESTINATION

- Separate “it runs” from “it runs usefully.”
- Match a model and a bounded task to available hardware.
- Verify output with a diff and an independent test.
- Draft a fictional world seed: World Bible v0.1.

The reframe

Week 2 opening — Your computer is more powerful than you think

The Reframe

Do not ask: "Can my computer run AI?" Ask: "What AI can my computer run well?"

Every AI-assisted task in this course sits at the intersection of five variables:

- **Hardware** — what you can run at all
- **Model** — what the loaded model is actually good at
- **Task size** — how much you asked for in one request
- **Context** — how much surrounding code or text the model has to hold
- **Verification** — how you will know whether the answer was right

A larger model does not rescue a task that is too big, too vague, or unverifiable. The machine is powerful. **You are the multiplier.** Slogan for the semester: love more, trust AI less.

Hardware vocabulary, a useful level

- **CPU** — general-purpose processor; always present; fine for small models and for orchestrating tools like Alder
- **RAM** — system memory; a model that will not fit in RAM cannot run on CPU alone
- **GPU** — a processor built for the parallel math models need
- **VRAM** — memory on the GPU itself; usually the real ceiling on which model sizes are practical, not raw GPU speed
- **Quantization** — compressing a model's numbers so it fits in less VRAM, at a small and usually acceptable cost in answer quality

"It runs" and "It runs usefully" are different claims.

Small models can be excellent at bounded work

A small local model asked to make one narrow, well-specified change to one file — and checked against a real test — is a reliable tool. The goal is the **smallest sufficient model for the bite**, not the largest model you can technically load. A larger model does not fix an underspecified or oversized request.

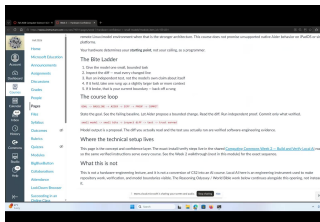
Local versus remote inference

Both are legitimate architecture choices, not a hierarchy:

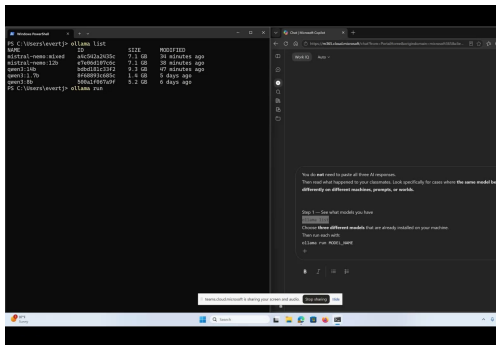
It runs *versus* it runs usefully.

The bite ladder

1. Give the model one small, bounded task.
2. Inspect the diff — read every changed line.
3. Run an independent test, not the model's claim.
4. If it holds, take one rung up: task or context.
5. If it breaks, you found a boundary; back off a rung.



See what models you actually have



The screenshot shows a Windows terminal window with the command `ollama list` executed. The output is a table of installed models. To the right, a web browser displays the Ollama website, which includes instructions on how to use the command.

NAME	ID	SIZE	MODIFIED
mistral-nemo:8.0	alc562a2405c	7.1 GB	34 minutes ago
mistral-nemo:12b	q7606a2f0ac	7.1 GB	35 minutes ago
qwen1.5:72b	b6b5d1c13f2	9.3 GB	47 minutes ago
qwen1.5:72b	8f46091c65c	1.4 GB	5 days ago
qwen1.5:72b	5b6a1607c0f	5.3 GB	6 days ago

PS C:\Users\evartj> ollama run

You do not need to paste all these AI responses. Then read what happened to your chatbot. Link specifically for cases where the same model behaved differently on different machines, prompts, or works.

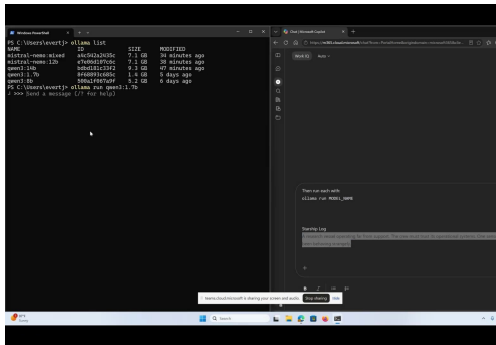
Step 1 — See what models you have

Choose three different models that are already installed on your machine. Then run each with:

```
ollama run <model_name>
```

- 'ollama list' inventories installed models.
- Size changes the fit, speed, and boundary.
- Choose the smallest sufficient model for the bite.

A World Bible is a design seed



```
PS C:\Users\evettj> ollama list
NAME                ID                SIZE    MODIFIED
-----
mistral-nemo:8.0    abc5d2d2d05c     7.1 GB  34 minutes ago
mistral-nemo:12b    9f6b6d1f0c       7.1 GB  35 minutes ago
qwen1.5             b0bd1d1c33f2     9.3 GB  47 minutes ago
qwen1.5:7b          8f40991c605c     1.6 GB  5 days ago
qwen1.5:72b         5b0d1d0f0c       5.1 GB  6 days ago
PS C:\Users\evettj> ollama run qwen1.5:7b
/ >>> Send a message (/ / For help)
```

- Choose Frontier Settlement, Investigation Bureau, Starship Log, or Small Business.
- Write a premise, cast, one flow, questions, unknowns, and object predictions.
- You are allowed to be wrong.

Mismatch is evidence

The image shows a Windows PowerShell terminal window on the left and a Microsoft Copilot chat window on the right. The PowerShell window displays the output of the command `ollama list`, showing a table of installed models. The Copilot chat window shows a conversation where the user asks for help with a sensor mismatch, and the AI provides a detailed troubleshooting guide.

NAME	ID	SIZE	MODIFIED
mistral-nemo:mixed	a4c542a2435c	7.1 GB	34 minutes ago
mistral-nemo:12b	e766d103f6c	7.1 GB	38 minutes ago
qwen3:1.7b	bdbd181c13f2	9.1 GB	47 minutes ago
qwen3:1.7b	8f68893c685c	1.4 GB	5 days ago
qwen3:1.7b	8b1a1f607a0f	5.2 GB	6 days ago

PowerShell Prompt:

```
PS C:\Users\levert> ollama run qwen3:1.7b
/ >>> A research vessel operating far from support. The crew must trust its operational systems. One sensor has been behaving strangely.
Thinking...
Okay, the user mentioned that one sensor in a system is behaving strangely. I need to figure out how to approach this. First, I should consider possible causes. Maybe the sensor is faulty, or there's an interference issue. Let me think about common sensor problems.

If the sensor is in a controlled environment, like a lab, maybe it's a calibration issue. If it's in a real-world setting, maybe environmental factors are affecting it. Also, signal noise could be a problem. I should suggest checking the sensor's calibration. If that's not working, maybe the sensor itself is faulty.

Another angle: could there be a software issue? Maybe the system is not properly interpreting the sensor data. Or maybe there's a communication problem between the sensor and the rest of the system. I should mention checking the communication protocols and data transmission.

Also, consider if the sensor is part of a larger system. Maybe there's a dependency issue. For example, if the sensor is used in a machine, a malfunction could affect other components. Maybe the user needs to check the system's overall health.

I should also think about diagnostic tools. Maybe the user can use a multimeter to check for voltage issues, or a logic analyzer to check the signal. If the sensor is a digital one, maybe checking the output levels. For analog sensors, checking the signal amplitude.

If the problem persists after checking these, maybe the sensor is defective. In that case, replacement or calibration might be needed. Also, maybe the user should consult the manufacturer's documentation for specific troubleshooting steps.
```

Copilot Chat:

Buddy, I think we just found the first teaching moment of the lab.

The assignment says:

"A research vessel operating far from support. The crew must trust its operational systems. One sensor has been behaving strangely."

That is not actually a troubleshooting request.

It's a World B&E prompt.

The model was supposed to help build a fictional setting. Instead, qwen3:1.7b interpreted it as a real-world engineering support ticket and started talking about oscilloscopes, PC calibration, and EMI.

That's evidence.

What I would do next

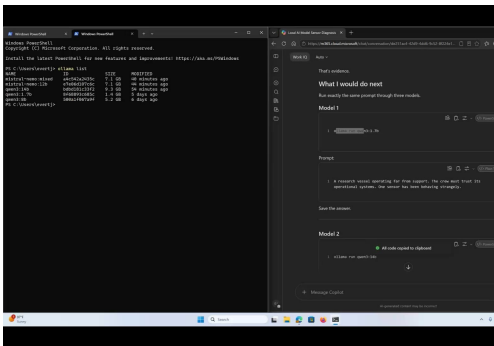
Run exactly the same prompt through these models.

Model 1

ollama run qwen3:1.7b

The fictional prompt became an engineering ticket. Diagnose the mismatch.

Fit is a memory question



Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

```
PS C:\Users\user> $PSVersionTable
```

Module	Version	Build	ReleaseDate	Path
Microsoft.PowerShell.Management	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Utility	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.CoreCLR.Diagnostics	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Security	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Host	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.LocalizedHelp	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Management	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Utility	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Diagnostics	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Security	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Management	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Utility	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Diagnostics	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Security	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\

What I would do next

Run exactly the same prompt through these models.

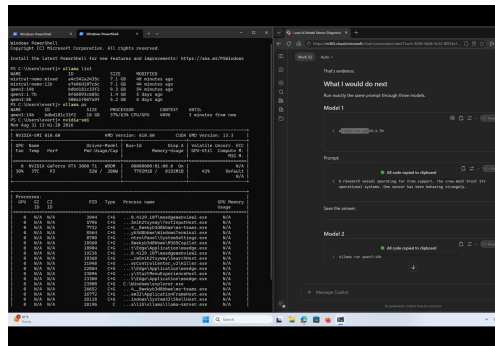
Model 1

Prompt

Save the session.

Model 2

All code copied to clipboard



Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

```
PS C:\Users\user> $PSVersionTable
```

Module	Version	Build	ReleaseDate	Path
Microsoft.PowerShell.Management	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Utility	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.CoreCLR.Diagnostics	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Security	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Host	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.LocalizedHelp	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Management	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Utility	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Diagnostics	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Security	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Management	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Utility	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Diagnostics	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\
Microsoft.PowerShell.Commands.Security	7.1.0	10	10/10/2023	C:\Program Files\PowerShell\7\

What I would do next

Run exactly the same prompt through these models.

Model 1

Prompt

Save the session.

Model 2

All code copied to clipboard

GPU-only is not always the answer

The screenshot shows a Windows PowerShell window and a web browser. The PowerShell window displays the output of the `slmgr /xpr` command, showing the Windows license expiration date as August 31, 2026. It also shows the output of the `wmic /namespace:\\root\\wmi /path:Wmi` command, which lists the available WMI classes. The web browser shows a page titled "Which is faster than" with a comparison of GPU-only vs. CPU+GPU performance. The page includes a table with the following data:

GPU	GPU Name	GPU Memory	GPU Usage	GPU Temperature
0	NVIDIA GeForce RTX 3090 Ti	24GB	100%	70C

The web browser also shows a comparison of GPU-only vs. CPU+GPU performance, with a table showing the following data:

GPU	GPU Name	GPU Memory	GPU Usage	GPU Temperature
0	NVIDIA GeForce RTX 3090 Ti	24GB	100%	70C

A split can make an oversized model run, but fit and speed are separate claims.

Before next class

RECAP

- “Runs usefully” requires a bounded task and evidence.
- Hardware is a starting point; VRAM is a real boundary.
- A model’s answer is a proposal until independently verified.
- Write World Bible v0.1 and leave its unknowns visible.